# Acids and Bases

Chapter
**8**

**1.** **[6 marks]** *3.2.4 (2016:31)*

(a) Select **one** basic, **one** acidic and **one** neutral salt from the list below to complete the table. [3]

KCN, $NH_4C\ell$, $Mg_3(PO_4)_2$, $NaNO_3$, $KHCO_3$, $NaCH_3COO$, $KC\ell$

| Acidic salt | Neutral salt | Basic salt |
|---|---|---|
| | | |

(b) Use the Brønsted-Lowry model to explain why the pH of ammonia solution is greater than 7.0 at 25 °C. Incorporate at least **one** appropriate equation in your answer. [3]

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**2.** **[6 marks]** *3.2.5 (2016:32)*

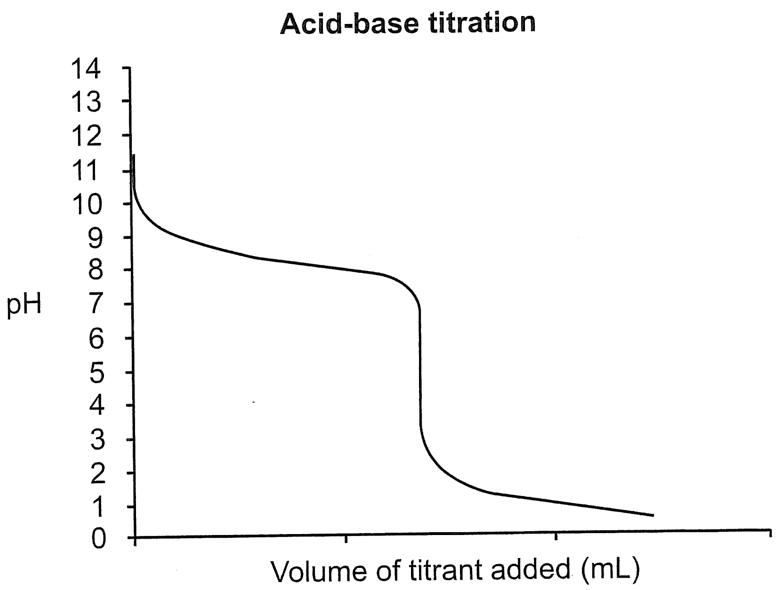
(a) A buffer of carbonic acid ($H_2CO_3$)/hydrogencarbonate ($HCO_3^-$) is present in blood plasma to maintain a pH between 7.35 and 7.45. Write an equation to show the relevant species present in a carbonic acid/hydrogencarbonate buffer solution. [2]

(b) Explain why 300.0 mL of 1.00 mol $L^{-1}$ carbonic acid/hydrogencarbonate buffer does **not** change in pH significantly when 3 drops of 1.00 mol $L^{-1}$ $HC\ell$ are added to it, yet when 3 drops of 1.00 mol $L^{-1}$ $HC\ell$ are added to 300.0 mL of distilled water there is a significant change in pH. [4]

**3.** **[6 marks]** *3.2.10, 3.2.9 (2016:34)*

The data below were collected from an acid-base titration.

(a) Label the equivalence point on the titration curve below using an arrow and record the pH value at this point. [2]




pH value at equivalence point: ____________________

(b) Select an indicator from the table below that would be **best** for this titration and justify your choice. [4]

| Indicator | Low pH colour | Transition pH range | High pH colour |
|---|---|---|---|
| Methyl Yellow | red | 2.1 – 3.3 | yellow |
| Bromocresol Green | yellow | 3.8 – 5.4 | blue |
| Bromothymol Blue | yellow | 6.0 – 7.6 | blue |
| Phenolphthalein | colourless | 8.3 – 10.0 | pink |
| Alizarin Yellow R | yellow | 10.2 – 12.0 | red |

Indicator: ____________________

Justification: ____________________

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**4.** **[8 marks]** *3.2.6 (2017:31)*

Water is capable of self-ionisation.

(a) Write an equation for the self-ionisation of water. [2]

(b) Write the equilibrium constant expression for the self-ionisation of water. [1]

(c) The equilibrium constant for the self-ionisation of water, Kw, is $1.00 \times 10^{-14}$ at 25 °C. What does this value indicate about this reaction? [1]

The K values for the self-ionisation of water at 100.0 kPa are given here for a number of different temperatures.

| Temperature (°C) | K value |
|---|---|
| 0 | $0.114 \times 10^{-14}$ |
| 25 | $1.00 \times 10^{-14}$ |
| 50 | $5.48 \times 10^{-14}$ |
| 75 | $19.9 \times 10^{-14}$ |
| 100 | $51.3 \times 10^{-14}$ |

(d) Calculate the pH of water at 50 °C. [2]

(e) Is water acidic, basic or neutral at 50 °C? State a reason for your answer. [2]

**5.** **[10 marks]** *3.2.10, 1.3 (2018:26)*

Solid copper(II) hydroxide is added to excess 0.100 mol $L^{-1}$ carbonic acid solution.

(a) Write the balanced equation, with appropriate state symbols, for the reaction that takes place between the copper(II) hydroxide and carbonic acid. [3]

(b) Predict **all** visible changes that would be observed, if any, while the reactants are mixed together and afterwards. [3]

(c) Predict **two** observations that would be different if excess 0.100 mol $L^{-1}$ hydrochloric acid was used instead of the 0.100 mol $L^{-1}$ carbonic acid. [2]

One: ____

Two: ____

(d) State **two** personal safety measures the experimenter should take when conducting these experiments. [2]

One: ____

Two: ____

# *Chapter 8: Acids and Bases*

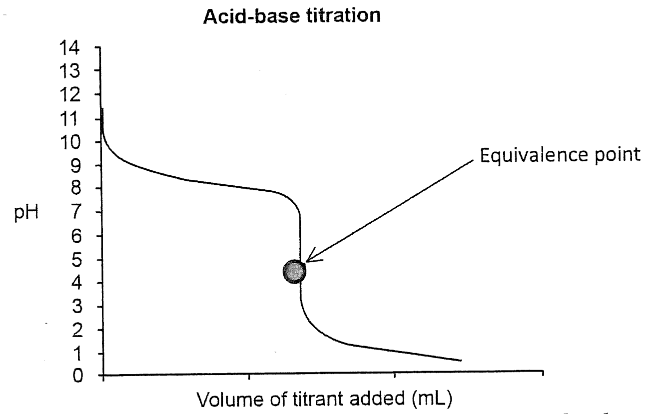
1.(2016:31) a)

| Acidic | Neutral | Basic |
|---|---|---|
| $NH_4C\ell$ | $KC\ell$, $NaNO_3$ | $KCN$, $Mg_3(PO_4)_2$ |

b) This question has two key parts you must address. It asks you to use Brønsted-Lowry theory (so $NH_3 + H^+ \rightarrow NH_4^+$ is not adequate) and to explain why the pH of ammonia solution is greater than 7.0 (you must explain the significance of these numbers and not just say it's basic).
Example:
$NH_3(g) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ – note the states are included and the equilibrium arrow is used.
Brønsted-Lowry discussion: Ammonia acts as a weak base accepting a proton from water, acting as an acid, to form the ammonium ion and hydroxide ions.
pH discussion: In water the concentration of $H^+$ and $OH^-$ are both $10^{-7}$ mol $L^{-1}$ This reaction means the concentration of $OH^-$ will increase above the $H^+$ so the pH will be above 7.

2.(2016:32) a) $H_2CO_3(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$ – note the equilibrium arrow.
b) This question tests your understanding of buffers. Most students know a buffer is a mixture of a weak acid (or base) and its conjugate in high concentration eg 1 mol $L^{-1}$.
In this question the $H_2CO_3/HCO_3^-$ meets this requirement. Here $HC\ell$ forms $H_3O^+$ in water
$HC\ell(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$
The $HCO_3^-$ accepts protons from $H_3O^+$ to form $H_2CO_3$ removing most of the newly added $H_3O^+$ions and so there is little change in the pH because the BUFFER CAPACITY has not been reached. But in the case of water the species are at $10^{-7}$ mol $L^{-1}$ concentration. Thus their BUFFER CAPACITY is small. Hence the very poor buffer is easily overwhelmed by the addition of the small quantity of $HC\ell$ which increases the concentration of $H_3O^+$ ions, lowering the pH.

3.(2016:34) a)




b) The syllabus does not mention any indicators you need to know specifically. This question requires you to know how to pick an indicator for a titration.
We have a weak base/strong acid titration which has an equivalence point at about pH 4.5. So we need an indicator with an end point of 4.5.
Bromocresol Green because ...

- Its colour change is at a suitable end point pH to show the equivalence point has been reached
- That a change in colour of the indicator (end point) indicates that the equivalence point has been reached
- Other indicators like Phenolphthalein, Alizarin Yellow R change too early while others like Methyl Yellow, Bromocresol Blue change too late and miss the equivalence point)
- Wrong indicator use will not be corrected by repeating the experiment therefore wrong indicator use is a systematic error

4.(2017:31) a) $2\,H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$
b) $K = [H_3O^+][OH^-]$
c) It tells us that there are very few ions produced. The ratio of reactant to product is huge. The constant says nothing about the rate of reaction.
d) At 50°C, $K = 5.48 \times 10^{-14}$
Since $K = [H^+] \times [OH^-]$ and they are equal then $[H^+] = \sqrt{5.48 \times 10^{-14}} = 2.34 \times 10^{-7}$
$pH = -\log 2.34 \times 10^{-7} = 6.63$
e) Water is neutral because $[H^+] = [OH^-]$ at all temperatures

5.(2018:26) a) As carbonic acid is weak we write it as $H_2CO_3(aq)$ not $H^+(aq)$ and $CO_3^{2-}(aq)$. And $Cu(OH)_2$ is insoluble.
$Cu(OH)_2(s) + H_2CO_3(aq) \rightarrow CuCO_3(s) + 2\,H_2O(\ell)$
b) A blue solid is mixed with a colourless, clear liquid. A green solid precipitate appears. $CuCO_3(s)$ is green).
c) The colourless solution would turn blue and there would be no solid remaining.
d) Select two from safety glasses, gloves, lab coats, closed shoes, tie back hair.

6.(2018:27) a) $H_2PO_4^-(aq) + H_2O(\ell) \rightleftharpoons HPO_4^{2-}(aq) + H_3O^+(aq)$.